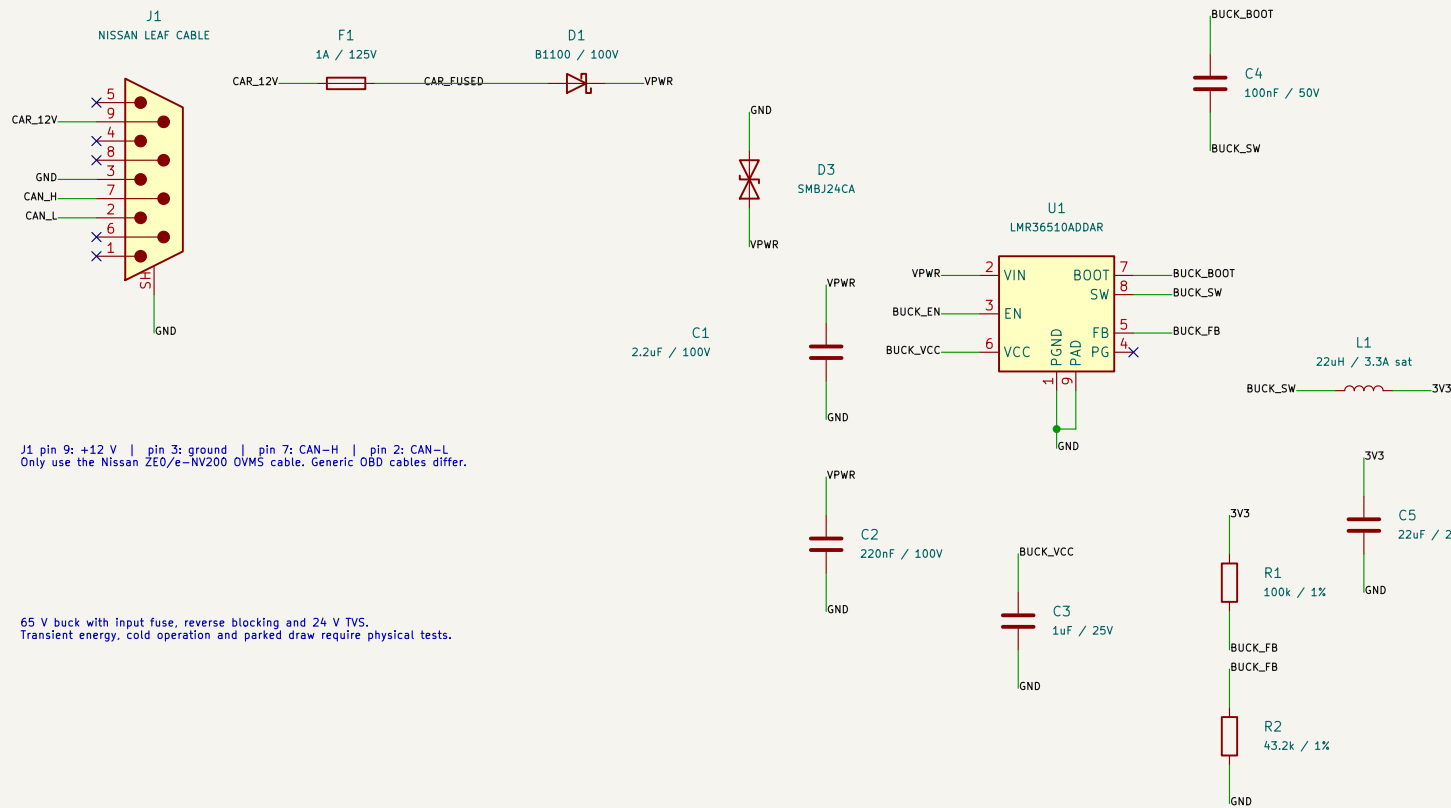


LEAF HEAT / Wi-Fi + one CAN channel

REV G.2 – HAND-SOLDER PROTOTYPE – FACTORY: U1 / U2 / U4 / U5 / U6 / L1

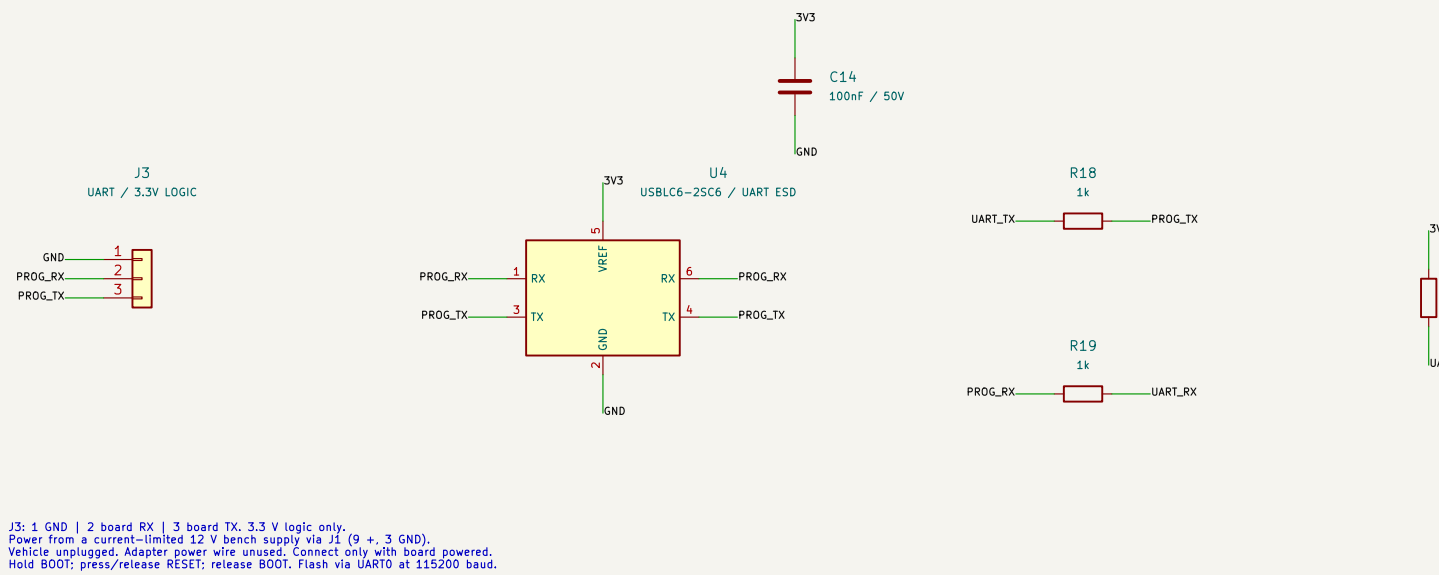
1 VEHICLE POWER / 3.3 V SUPPLY



J1 pin 9: +12 V | pin 3: ground | pin 7: CAN-H | pin 2: CAN-L
Only use the Nissan ZEO/e-NV200 DVMS cable. Generic OBD cables differ.

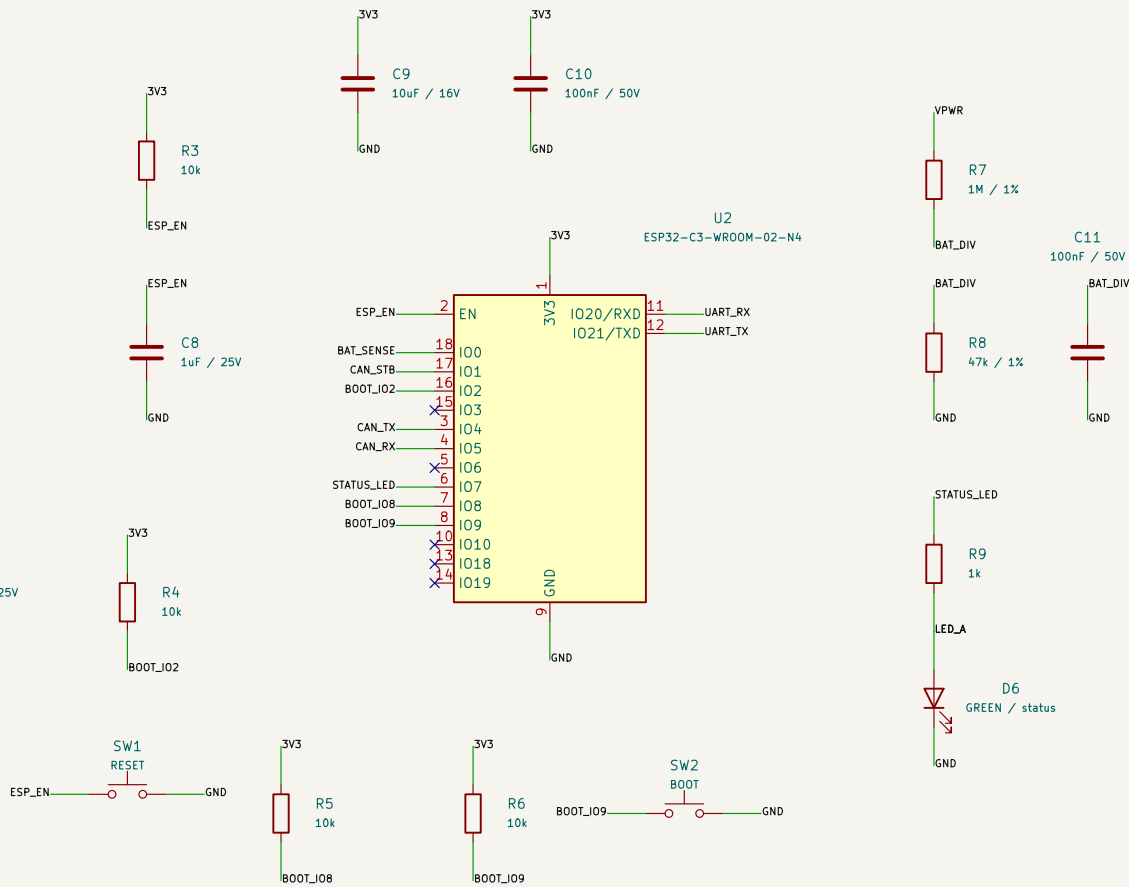
65 V buck with input fuse, reverse blocking and 24 V TVS.
Transient energy, cold operation and parked draw require physical tests.

3 UART PROGRAMMING – NO USB / NO PROGRAMMER POWER

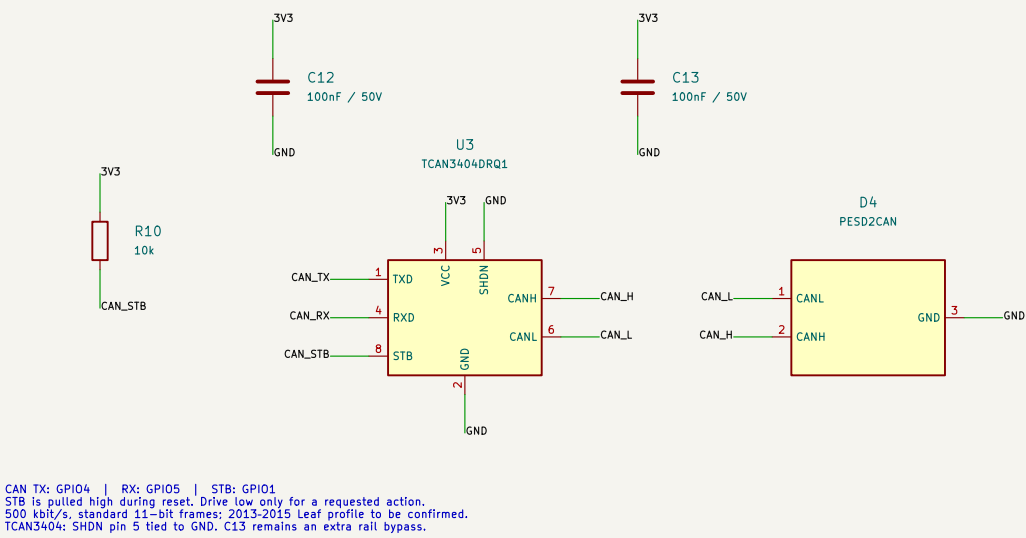


J3: 1 GND | 2 board RX | 3 board TX. 3.3 V logic only.
Power from a current-limited 12 V bench supply via J1 (9 + 3 GND).
Vehicle unplugged. Adapter power wire unused. Connect only with board powered.
Hold BOOT: press/release RESET; release BOOT. Flash via UART0 at 115200 baud.

2 Wi-Fi / BOOT / BATTERY SENSE



4 VEHICLE CAN – NO TERMINATION

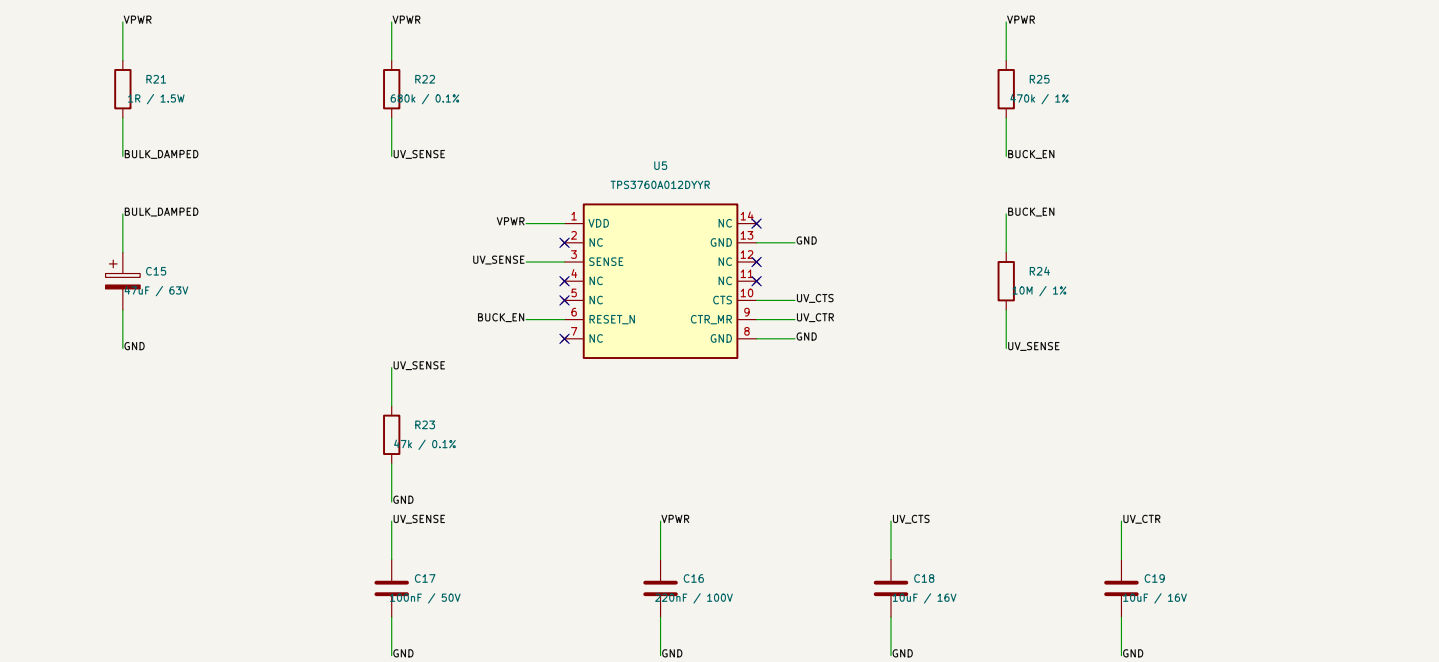


CAN TX: GPIO4 | RX: GPIO5 | STB: GPIO1
STB is pulled high during reset. Drive low only for a requested action.
500 kbit/s, standard 11-bit frames: 2013-2015 Leaf profile to be confirmed.
TCAN3404: SHDN pin 5 tied to GND. C13 remains an extra rail bypass.

Rev G.2: Independent UV monitor controls buck EN; no scheduled Wi-Fi disconnect.
Voltage measurement is after D1 and includes R26; calibrate on the bench.

GPIO3 / GPIO18 / GPIO19 unused. No USB-presence sensing in Rev G.2.
OTA updates need firmware implementation, UART remains the recovery interface.

5 INPUT DAMPING / INDEPENDENT BATTERY PROTECTION



UV monitor is non-latching. Low battery disables the regulator, including Wi-Fi.
Delay and hysteresis prevent rapid restart cycling. See protection calculations.

Factory fits U1 U2 U4 U5 U6 L1. All other parts hand fitted.
Prototype bench, assembly-process and vehicle validation required.

Engineering review and vehicle tests required before release

DIY Leaf project

Sheet: /
File: leaf-heat-v7.kicad_sch

Title: Leaf Heat – Wi-Fi climate control

Size: A2 Date: 2026-09-13 Rev: G.2
KiCad E.D.A. 10.0.6 Id: 1/1